CLUB MEETING

Servers

What they are, how they work, and how to run your own

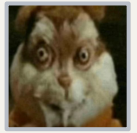
[Student Integrated Technologies] · [By club pres]



What is a server?

A server is just a computer whose job is to serve other computers.

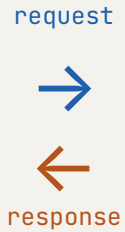
Files, websites, games, data. Any computer can be a server; what makes it one is the role it plays, not the hardware.



How it works: request and response

Client

Your laptop, phone or game console. It asks for something.



Server

Always listening. Does the work and sends back an answer.

Example:

1.) you type a URL
2.) your browser asks a web server for the page
3.) the server sends back the HTML
4.) your browser draws/displays it.

Kinds of servers

Web
Sends websites to browsers

File
Stores and shares files across a network (NAS)

Database
Holds data that apps look up

Game
Keeps every player in the same world (Minecraft)

Media
Streams your movies and music (Jellyfin, Plex)

Many more
basically so much things servers can do.

Server hardware vs. your PC

	Your PC or you home computer	Typical server somewhere in data centers	A homelab usually at someone's home/office
Shape	Tower, laptop, table, or phone	Rack-mounted (1U, 2U...)	Tower, laptop, tablet, mini pc, rack-mounted, anything
Uptime	When you use it	mostly running 24/7	when you want to, or 24/7
Memory	Regular RAM	ECC RAM that corrects bit errors	Regular ram or ecc(recommended)
Power	One power supply	Redundant power supplies	one or more power supplies
Storage	One or two drives	Many drives in RAID	One or as much as you need
Control	Monitor and keyboard	Headless, managed remotely	both remote, or directly

The software side

Operating system

Linux runs most servers: Ubuntu Server, Debian. Usually no desktop, just a command line you reach over SSH.

Services and ports

Programs that listen on a port number: web on 80/443, SSH on 22, Minecraft on 25565.

Virtualization

One machine, many servers. Virtual machines (Proxmox) and containers (Docker) keep things separate.

Where do servers live?

On-premises

Racks in a company's or school's own building. Full control, full responsibility.

Cloud

Rent someone else's servers by the hour: AWS, Google Cloud.

Homelab

A few machines at home to learn and tinker on. The best way to start.

Run your own in four steps

1 Grab hardware
An old laptop/computer, a Raspberry Pi, or a cloud free tier

2 Install Linux
Ubuntu Server is free and beginner-friendly

3 Connect over SSH
Control it from your own computer's terminal by ~ssh username@device_ip

4 Host something
A Minecraft server, a website, Pi-hole, jellyfin, etc

Questions?

please ask.



Thank you!

Qr codes :D

Insta (old)

Discord

Github